



RV College of Engineering®

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TABLE OF CONTENTS

Sl.no	Section	Page Number
1.	Executive Summary	
2.	Introduction	
3.	Activities and Events	
4.	Membership and Financial Overview	
5.	Achievements and Recognition	
6.	Challenges	
7.	Future Plans	
8.	Acknowledgements, Appendices and Contact Information	



EXECUTIVE SUMMARY

The **SPARK – IUCEE RVCE Student Chapter** continues to serve as a platform for innovation, technical learning, and socially responsible engineering. In the academic year **2025–26**, the chapter focused on bridging classroom knowledge with hands-on experience, fostering creativity, collaboration, and holistic development among students.

Key technical initiatives included **Razz Security (Cybersecurity Workshop)**, **AI in Engineering**, and the **Sensors & Processor Requirements workshop for ML, and Quantum Computing**, which enhanced system-level understanding, collaborative problem-solving, and exposure to emerging technologies. Programs such as the **Induction Program** and **Investiture Ceremony** strengthened leadership continuity and ensured smooth onboarding of new members.

Experiential learning was emphasized through industrial visits to the **Kaiga Nuclear Power Plant** and the **Integral Coach Factory**, providing firsthand exposure to large-scale engineering operations and real-world industrial workflows. Outreach initiatives with the **Agastya Foundation** and programs like **Shakti Shiksha** fostered STEM curiosity among school students and encouraged meaningful community engagement.

The chapter also launched **“The Insight” podcast series**, which bridged faculty and students, sharing professional journeys, research perspectives, and club initiatives to inspire and guide student growth.

Notable achievements included awards at **Aavishkar Manthan**, **AUNSF 3.0**, **Ambition National Hackathon**, and recognition as the **Best Student Chapter at IASF 2025**. Through these efforts, SPARK continues to nurture engineers who are technically proficient, socially conscious, and ethically grounded, equipped to contribute meaningfully at both national and global levels.



INTRODUCTION

SPARK: Study through Projects and Activities for Renewing Knowledge

SPARK aims to bridge the gap between theoretical knowledge and practical application, empowering students to innovate, collaborate, and grow into socially responsible engineers. Through diverse projects, activities, and learning experiences, SPARK nurtures creativity and develops well-rounded professionals ready to address global challenges.

VISION

Believe in studying through projects and activities for renewing knowledge.

MISSION

- Aim to enrich and enhance our community through technical and cultural diversity, curiosity, and the overall development of engineering students.
- Improving the educational experience beyond classroom walls and enhancing our ethical values.
- Innovations by out-of-the-box thinking.

Goals:

- Promote interdisciplinary collaboration
- Develop professional skills
- Build platforms for real-world innovation
- Ensure social responsibility in engineering

SPARK is a student-led chapter that blends technical skills with soft skills, industry insights, and personal growth opportunities to build engineers who think creatively. Emphasizing diversity and collaboration, SPARK unites students from various disciplines and backgrounds, providing platforms for engaging events through research projects and industry visits.

By fostering spaces where potential is nurtured and celebrated, SPARK aligns with RV College's commitment to holistic development. Through real-world, experiential learning, SPARK empowers students to become leaders who solve complex problems, drive innovation, and create meaningful societal impact.



ACTIVITIES AND EVENTS CONDUCTED IN THE ACADEMIC YEAR 2025-2026

WORKSHOPS & SEMINARS

27-12-24	Razz Security – Cybersecurity Workshop	A hands-on cybersecurity workshop was organized under the SPARK IUCEE RVCE Student Chapter. The session provided students with practical knowledge on identifying, preventing, and mitigating cyber threats. Participants gained awareness of data security, ethical hacking, and safe digital practices, making the session highly insightful and impactful.
17-10-25	AI in Engineering – Industry Expert Session	A speaker session on AI in Engineering was organized under the SPARK – IUCEE RVCE Student Chapter. The session was delivered by Ms. Girija Kolagada, Vice President of Engineering at Progress Chef, who shared valuable insights on applying Artificial Intelligence in real-world engineering scenarios. Participants gained awareness of AI techniques, data analytics, and automation trends, making the session highly insightful and impactful.
22-11-25, 29-11-25	Sensor and Processor Requirements for AIML and Quantum Computing	A workshop on Sensor and Processor Requirements for AIML and Quantum Computing was organized under the SPARK – IUCEE RVCE Student Chapter. The event focused on emerging hardware requirements for AI, Machine Learning, and Quantum Computing, highlighting sensor technologies, processor architectures, and system-level design considerations. Participants gained insights into classical-to-quantum representations, optimization mechanisms, and quantum-aware hardware design, making the workshop highly informative and impactful.



20-12-25	Design Fest	A Design Fest was organized under the SPARK – IUCEE RVCE Student Chapter as part of the AIML and Quantum Computing workshop series. The event provided participants with a platform to apply their technical knowledge in developing innovative hardware-centric and system-level design solutions. Through project presentations and evaluations, students enhanced their understanding of real-world engineering challenges, making the event engaging and impactful.
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COLLABORATIVE PROJECTS

30-08-25 & 06-09-25	Agastya Foundation Collaboration Event	In collaboration with the Agastya Foundation, SPARK students conducted outreach sessions in government schools. The event focused on teaching science and mathematics concepts in a simple, engaging manner, fostering curiosity and critical thinking among school children. The collaboration highlighted the importance of student-led social innovation and community impact.
23-10-2025	Shakti Shiksha	The Shakti Shiksha initiative was conducted as a social outreach program at Sri Jnanakshi Vidyaniketan School. The team visited the school and actively interacted with students through engaging discussions and activities aimed at building self-confidence and self-awareness. The sessions emphasized the importance of strong, respectful relationships, personal well-being, and mutual respect, creating a safe and supportive environment. The initiative fostered positive learning experiences and contributed meaningfully to the students' personal and social development.

EVENTS CONDUCTED BY THE SPARK STUDENT CHAPTER



22-04-25	Investiture Ceremony of the Current Core Team	The formal investiture ceremony marked the beginning of the tenure of the new core team of the SPARK IUCEE RVCE Student Chapter. The event highlighted the achievements of the outgoing team while officially passing responsibilities to the new members. The ceremony fostered leadership, responsibility, and commitment within the chapter.
16-05-25 - 18-05-25	EduAlthon	EduAlthon, was conducted to explore innovative intersections of Artificial Intelligence and Education. Over 50+ student teams participated, presenting novel ideas and prototypes. The event encouraged problem-solving, teamwork, and innovation in leveraging AI for educational advancement.
28-08-25	Induction Program for the Incoming Batch of 2025	The induction program was organized to welcome and orient the newly admitted students. It featured a series of four events covering multiple engineering domains, designed to foster critical thinking, problem-solving, and interactive learning. The sessions provided participants with a comprehensive introduction to technical concepts while encouraging collaboration and analytical skills.
05-12-25	Mystery.EXE	The Mystery.EXE event was organized as a technical treasure hunt as part of the college fest 8th Mile. The event featured multiple rounds involving technical clues, logical puzzles, and problem-solving challenges, encouraging participants to apply engineering concepts in a fun and competitive setting. It promoted teamwork, critical thinking, and analytical skills, making the event engaging and intellectually stimulating.

INDUSTRY VISITS

15-03-25	Kaiga Nuclear Power Plant Visit	Students from the SPARK chapter visited the Kaiga Nuclear Power Plant, gaining firsthand exposure to nuclear energy generation, reactor operations, and safety measures. The visit provided a deeper understanding of energy systems and India's progress in nuclear power technology.
09-12-25	Integral Coach Factory, Chennai	Students from the SPARK – IUCEE RVCE Student Chapter visited the Integral Coach Factory (ICF), gaining firsthand exposure to railway coach manufacturing processes, advanced fabrication techniques, and quality assurance practices. The visit provided valuable insights into large-scale industrial operations, engineering precision, and India's advancements in railway manufacturing technology.

THE INSIGHT – PODCAST BY SPARK

“The Insight” is an exclusive podcast series initiated by the SPARK – IUCEE RVCE Student Chapter, conceptualized to bridge the gap between faculty and students at RV College of Engineering. The series serves as a platform where faculty members share their professional journeys, research interests, teaching philosophies, and personal experiences, helping students connect with them beyond the classroom environment.

What started as a medium to foster stronger communication between students and faculty has now expanded into a dynamic initiative that also features various innovative student clubs across the campus. By highlighting the vision, activities, and impact of these clubs, *The Insight* enables students to discover the diverse opportunities available within RVCE and make informed choices that align with their skills and passions.

So far, insightful episodes have been recorded with the Principal, Vice Principal, Faculty Advisor (former Dean Academics), and the Dean – Cultural, each sharing inspiring perspectives on education, leadership, and student growth. In addition, multiple podcast episodes featuring technical and social clubs have been successfully recorded. Several other sessions are



currently in the editing and production pipeline, as the series continues to evolve into a recognized storytelling platform celebrating the people, ideas, and initiatives shaping the RVCE community.

MEMBERSHIP OVERVIEW

Category	Number of Members	Roles / Breakdown
Chairman	1	Dr. K N Subramanya, Principal
Faculty Advisors	2	Dr. K.S. Geetha , Vice Principal Dr. Shanmukha Nagaraj, HoD-ME
Faculty Coordinator	1	Dr. Prapulla S B, Associate Professor
Core Team	5	President, Vice President, Secretary, Joint Secretary, Treasurer
Wings	9	Projects Research and Development, Industrial Connect and People Operations, Activities and Events, Social Media, Women Empowerment
Members	103	31 - Prefects 72 - Members

FINANCIAL OVERVIEW

A significant portion of the chapter's income was generated through entry fees collected from various activities, including industry visits to Kaiga Nuclear Power Plant and ICF Chennai. Certain flagship events, such as Aavishkar Manthan (₹19,200) and Sensor and Processor Requirements for ML and Quantum Computing (₹15,000), were conducted using college-sanctioned budgets. Participation in external forums and conferences was partially supported by institutional funding, with the remaining expenses borne by the participating students.

In addition, the SDG – Youth Photography Event (₹5,000) contributed to revenue generation through participant registration fees, while also allocating a prize pool for the top three rank holders, thereby balancing engagement with incentive-based recognition.

Major Sources of Funding

The chapter's primary sources of funding included event entry fees, college-approved financial support for key programs, and self-financing by students for select activities such as external visits and forums.

Budget Analysis and Financial Health

The chapter maintained a sound financial position throughout the year, characterized by efficient fund utilization and responsible expenditure planning. A healthy balance between income and expenses was consistently upheld, enabling the successful execution of planned activities while ensuring long-term financial sustainability.

CHALLENGES

The challenges faced in the previous year were largely overcome by gaining hands-on experience in organizing and managing college-wide initiatives, particularly through **EduAithon**, which enhanced our ability to plan and execute large-scale events. We also initiated inter-club collaborations to promote interdisciplinary learning, beginning with **Anoraniya**, the Quantum Club, through which a workshop on Sensor and Processor Requirements for AIML, and Quantum Computing was successfully conducted. Despite this progress, the following challenges were encountered during the year:

- Variations in academic calendars across different academic years led to overlapping examination schedules, resulting in uneven student availability and difficulties in conducting events and maintaining consistent project monitoring.
- Toward the end of the semester, the overlap of academic commitments such as Experiential Learning projects and end-semester examinations increased workload pressures, making it challenging to balance academic responsibilities with chapter activities.
- Sustaining interest among students from diverse courses and domains remains a gradual process, as engagement with emerging technologies often requires broader visibility and time to gain consistent participation

FUTURE PLANS

- The student chapter aims to evolve into a comprehensive platform for **deep technical learning, interdisciplinary collaboration, and socially responsible engineering**.
- Building on initiatives such as the *Sensors and Processor Requirements* workshop, the chapter plans to strengthen **system-level engineering expertise**, including sensor selection, signal conditioning, processor architecture, mechanical integration, and real-world design trade-offs.
- Future efforts will focus on developing **end-to-end engineering solutions** by integrating hardware, software, algorithms, and mechanical design, moving beyond isolated or domain-specific implementations.
- Technical engagement will be enhanced through **structured learning tracks, hands-on workshops, technical deep dives, long-term**

projects, and industry visits, providing exposure to real-world engineering practices and industrial workflows.

- In addition to technical growth, the chapter envisions fostering **holistic student development** through initiatives such as *Shakti Shiksha*, photography competitions, community outreach programs, and interactions with industry professionals.
- These activities aim to cultivate **social awareness, leadership, communication skills, and ethical responsibility**, ensuring the development of engineers who are technically competent, socially conscious, and capable of meaningful contributions to society.

AWARDS & RECOGNITIONS RECEIVED BY MEMBERS

Event / Program	Achievement	Members Involved
Aavishkar Manthan	1st Place – Elocution	Prajwal J, Rupini H, Kavini Krishan, Dakshraj
	1st Place – Ideathon	Monish G, Parinitha
	3rd Place – Ideathon	Vijayalaxmi Ashok Patil, Yashwanth S, Chandana B C, A S S S Koundinya
AUNSF 3.0	Special Mentions (All Themes)	Yashwanth R, Chinmayi Devi H N, Dakshraj Shetty, Manish Kumar C S, Yashwanth S, Ashok R
Ambition National Hackathon	First Prize (among 128 teams)	Kavini Krishnan
BIT National Project Expo	Project Exhibition	Kavini Krishnan, Chandana B C, Ameya Acharya
HDSE Leadership Workshop (Mysore)	Participation	Kavini Krishnan, Monish Gangadharan
Srujana Science Exhibition (JSS Academy, Mysore)	First Place	Kavini Krishnan, Ameya Acharya, Yashwanth S, Dakshraj, Manish
IEEE Mini Project Symposium	Project Presentation	Kavini Krishnan, Monish Gangadharan, Chinmayi

(ATME College)		Devi
IASF 2025 (VNRVJIET)	Best Student Chapter – IUCEE	SPARK Chapter, RVCE

RESEARCH PAPERS & PUBLICATIONS

- “*Aerodynamic Wing Optimization Using Deep Reinforcement Learning: A Comparative Study of PPO and DQN*” by **Ashok R and Tarun Prabhash** was presented at the **2025 9th International Conference on Computational System and Information Technology for Sustainable Solutions (CSITSS)**, organized by IEEE and published in IEEE Xplore.

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- Quiklrn

Their guidance, cooperation, and support have been integral to our success.